Assignment 01 Solution Document

Course: Cloud Computing (CC-F26)
Section: BSE-V(A)
Instructor: Prof. Waqas Saleem
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STUDENT INFORMATION

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- Section: BSE-V(A)
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Task 1: Setting Up Gitea with Docker on Ubuntu Server

Step 1.1: Gitea Container Setup

Description: Verified that the Gitea container is up and running in Docker on the local Ubuntu Server environment.*(Insert Screenshot: task1_gitea_running.png)*

```
kashafnoor22@ubuntu:~/Gitea$ docker compose ps
NAME       IMAGE               COMMAND                  SERVICE   CREATED         STATUS         PORTS
gitea      gitea/gitea:latest  "/usr/bin/entrypoint…"   gitea     30 minutes ago  Up 30 minutes  22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db   postgres:17-alpine  "docker-entrypoint.s…"   db        30 minutes ago  Up 30 minutes  5432/tcp
kashafnoor22@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
kashafnoor22@ubuntu:~/Gitea$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:1e:40:d7 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c291e40d7
    inet 192.168.205.129/24 metric 100 brd 192.168.205.255 scope global dynamic ens33
       valid_lft 1020sec preferred_lft 1020sec
    inet6 fe80::20c:29ff:fe1e:40d7/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 62:bf:de:fc:fb:fe brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
4: br-85364d60de32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 5e:88:f9:ee:61:ec brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-85364d60de32
       valid_lft forever preferred_lft forever
    inet6 fe80::5c88:f9ff:feee:61ec/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
5: vethc40d508@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-85364d60de32 state UP group default
    link/ether 02:a7:1b:4c:cc:92 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::a7:1bff:fe4c:cc92/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
6: veth4c7e12a@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-85364d60de32 state UP group default
    link/ether 56:7d:e5:08:17:de brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::5cc8:81ff:fe99:fd4c/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
kashafnoor22@ubuntu:~/Gitea$ _
```

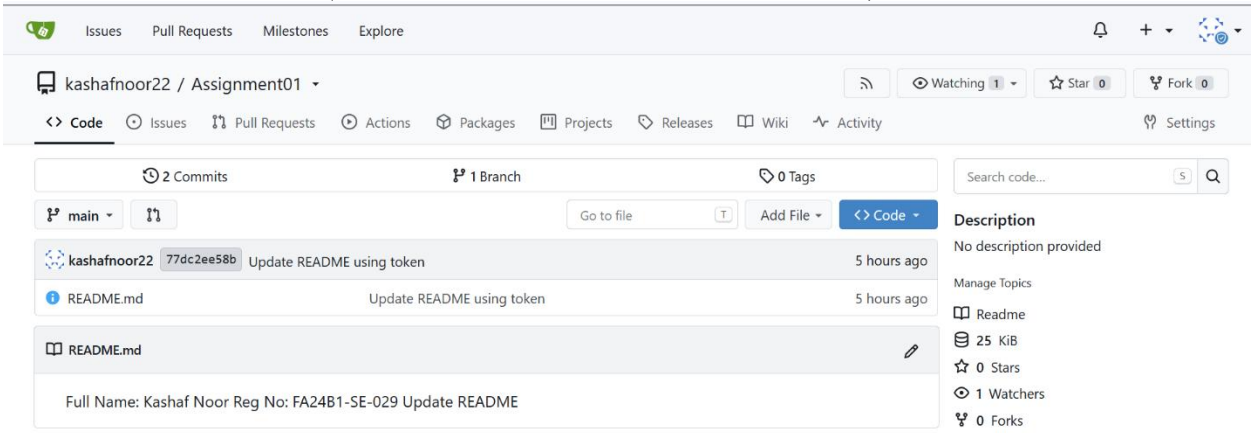
Step 1.2: First Push to Gitea

Description: Initialized the local repository and successfully performed the first git push to the Gitea remote instance.*(Insert Screenshot: task1_gitea_push.png)*

```
kashafnoor22@ubuntu:~$ cd ~/Gitea
kashafnoor22@ubuntu:~/Gitea$ docker compose up -d
[+] up 2/2
 V Container gitea    Running                                                                                    0.0s
 V Container gitea_db Running                                                                                    0.0s
kashafnoor22@ubuntu:~/Gitea$ docker compose ps
NAME       IMAGE              COMMAND                  SERVICE   CREATED      STATUS         PORTS
gitea      gitea/gitea:latest   "/usr/bin/entrypoint…"   gitea     6 hours ago   Up 2 minutes   22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db   postgres:17-alpine   "docker-entrypoint.s…"   db        6 hours ago   Up 2 minutes   5432/tcp
kashafnoor22@ubuntu:~/Gitea$ cd ~/Assignment01
kashafnoor22@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://kashafnoor22@127.0.0.1:3000':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 315 bytes | 63.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/kashafnoor22/Assignment01.git
   71b412e..77dc2ee  main -> main
branch 'main' set up to track 'gitea/main'.
kashafnoor22@ubuntu:~/Assignment01$
```

Step 1.3: Gitea Repository Web View

Description: Checked the Gitea web interface showing the pushed Assignment01 repository along with the README.md file.*(Insert Screenshot: task1_gitea_repository.png)*



Task 2: Dual Remote Configuration & GitHub Integration

Step 2.1: Dual Remotes Setup

Description: Configured two remotes (gitea and github) in the local Git configuration using git remote -v.*(Insert Screenshot: task2_remotes.png)*

```
kashafnoor22@ubuntu:~/Assignment01$ cd ~/Assignment01
kashafnoor22@ubuntu:~/Assignment01$ git remote add github https://github.com/kashafnoor22/Assignment01.git
kashafnoor22@ubuntu:~/Assignment01$ git remote -v
gitea   http://kashafnoor22@127.0.0.1:3000/kashafnoor22/Assignment01.git (fetch)
gitea   http://kashafnoor22@127.0.0.1:3000/kashafnoor22/Assignment01.git (push)
github  https://github.com/kashafnoor22/Assignment01.git (fetch)
github  https://github.com/kashafnoor22/Assignment01.git (push)
kashafnoor22@ubuntu:~/Assignment01$ _
```

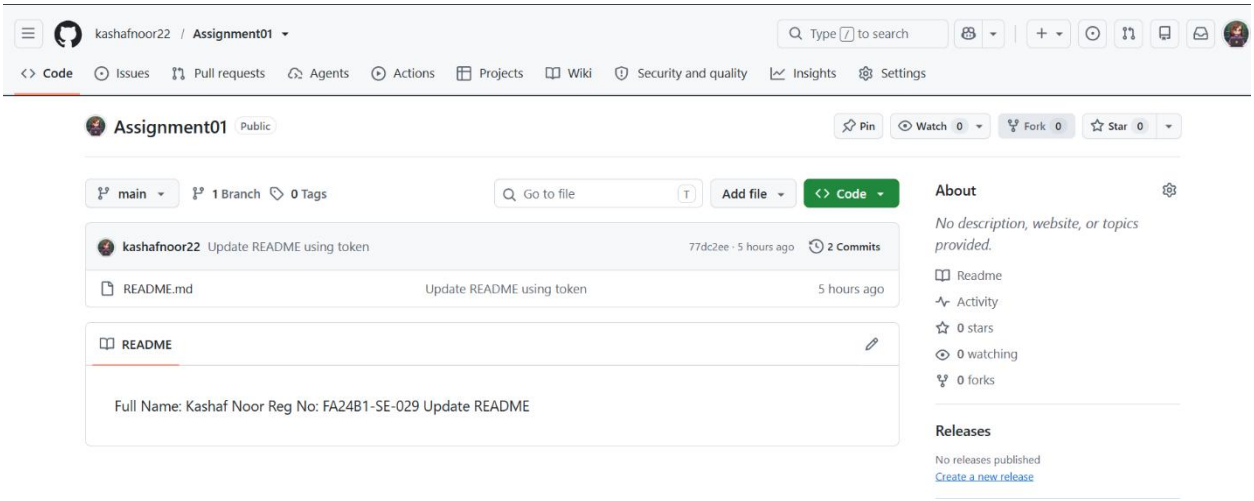
Step 2.2: Code Sync to GitHub

Description: Successfully pushed the repository source code to the GitHub remote repository using the Personal Access Token.*(Insert Screenshot: task2_github_push.png)*

```
kashafnoor22@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': kashafnoor22
Password for 'https://kashafnoor22@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 544 bytes | 181.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kashafnoor22/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
kashafnoor22@ubuntu:~/Assignment01$
```

Step 2.3: GitHub Repository Web View

Description: Verified the pushed repository files on GitHub's browser interface.*(Insert Screenshot: task2_github_repository.png)*



Task 3: Managing Binary Files with Git LFS

Step 3.1: Git LFS Initialization

Description: Installed Git LFS on Ubuntu Server, initialized LFS tracking for *.bin files, and verified .gitattributes.*(Insert Screenshot: task3_lfs_setup.png)*

```
kashafnoor22@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
28 packages can be upgraded. Run 'apt list --upgradable' to see them.
kashafnoor22@ubuntu:~$ sudo apt install git-lfs -y
Installing:
  git-lfs

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 28
  Download size: 4,309 kB
  Space needed: 13.1 MB / 3,235 MB available

Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/universe amd64 git-lfs amd64 3.7.1-1 [4,309 kB]
Fetched 4,309 kB in 19s (224 kB/s)
Selecting previously unselected package git-lfs.
(Reading database… 139923 files and directories currently installed.)
Preparing to unpack …/git-lfs_3.7.1-1_amd64.deb…
Unpacking git-lfs (3.7.1-1)…
Setting up git-lfs (3.7.1-1)…
Processing triggers for man-db (2.13.1-1build1)…
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kashafnoor22@ubuntu:~$ cd ~/Assignment01
kashafnoor22@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
kashafnoor22@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
kashafnoor22@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
kashafnoor22@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
kashafnoor22@ubuntu:~/Assignment01$
```

Step 3.2: Tracking Large Files

Description: Created three 10MB binary files (file1.bin, file2.bin, file3.bin) and verified tracking with git lfs ls-files.*(Insert Screenshot: task3_lfs_files.png)*

```
kashafnoor22@ubuntu:~/Assignment01$ dd if=/dev/urandom of=file1.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0447775 s, 238 MB/s
kashafnoor22@ubuntu:~/Assignment01$ dd if=/dev/urandom of=file2.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.046739 s, 228 MB/s
kashafnoor22@ubuntu:~/Assignment01$ dd if=/dev/urandom of=file3.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0451927 s, 233 MB/s
kashafnoor22@ubuntu:~/Assignment01$ ls -lh *.bin
-rw-rw-r-- 1 kashafnoor22 kashafnoor22 10M Oct  5 18:46 file1.bin
-rw-rw-r-- 1 kashafnoor22 kashafnoor22 10M Oct  5 18:46 file2.bin
-rw-rw-r-- 1 kashafnoor22 kashafnoor22 10M Oct  5 18:47 file3.bin
kashafnoor22@ubuntu:~/Assignment01$ git add .gitattributes file1.bin file2.bin file3.bin
kashafnoor22@ubuntu:~/Assignment01$ git commit -m "Add large binary files tracked with Git LFS"
[main ef51110] Add large binary files tracked with Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 file1.bin
 create mode 100644 file2.bin
 create mode 100644 file3.bin
kashafnoor22@ubuntu:~/Assignment01$ git lfs ls-files
56e7a69519 * file1.bin
2a8a92e67e * file2.bin
673d624e6a * file3.bin
kashafnoor22@ubuntu:~/Assignment01$
```

Step 3.3: LFS Object Push

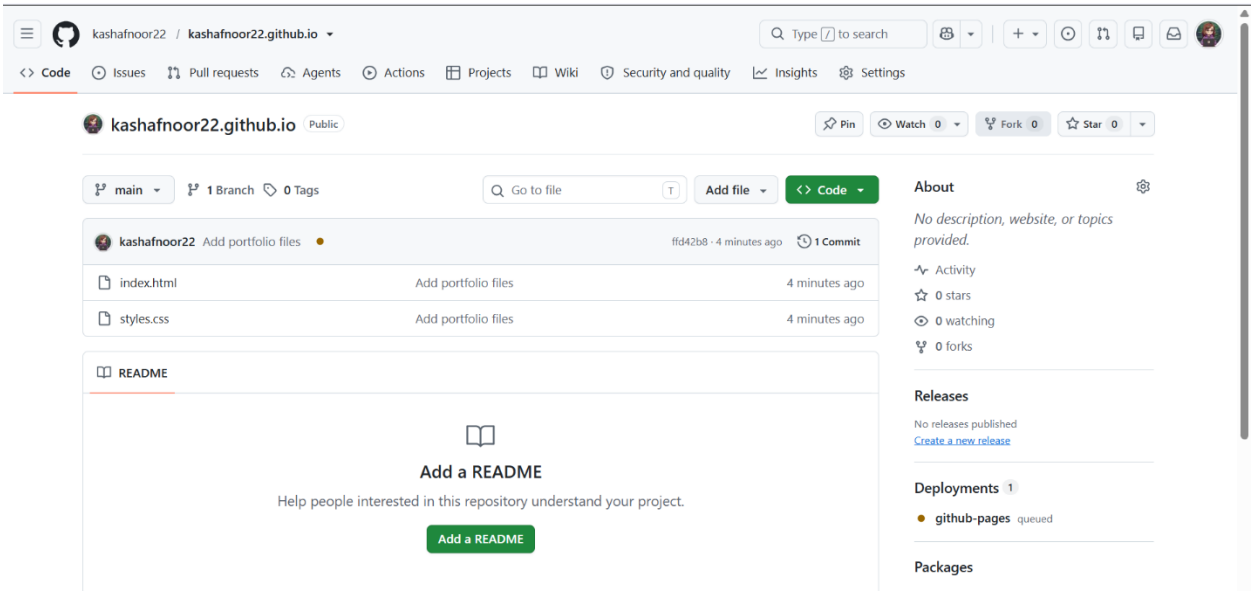
Description: Successfully uploaded and pushed the Git LFS-tracked binary objects to both Gitea and GitHub remotes.*(Insert Screenshot: task3_lfs_push.png)*

```
kashafnoor22@ubuntu:~/Assignment01$ git push gitea main
Password for 'http://kashafnoor22@127.0.0.1:3000':
Password for 'http://kashafnoor22@127.0.0.1:3000':
Locking support detected on remote "gitea". Consider enabling it with:
  $ git config lfs.http://kashafnoor22@127.0.0.1:3000/kashafnoor22/Assignment01.git/info/lfs.locksverify true
Uploading LFS objects: 100% (3/3), 32 MB | 0 B/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 795 bytes | 795.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/kashafnoor22/Assignment01.git
   77dc2ee..ef51110  main -> main
kashafnoor22@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': kashafnoor22
Password for 'https://kashafnoor22@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/kashafnoor22/Assignment01.git/'
kashafnoor22@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': kashafnoor22
Password for 'https://kashafnoor22@github.com':
Username for 'https://github.com': kashafnoor22
Password for 'https://kashafnoor22@github.com':
Username for 'https://github.com': kashafnoor22
Password for 'https://kashafnoor22@github.com':
Enumerating objects: 7, done.3/3), 32 MB | 852 KB/s
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 795 bytes | 265.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kashafnoor22/Assignment01.git
   77dc2ee..ef51110  main -> main
kashafnoor22@ubuntu:~/Assignment01$ _
```

Task 4: Portfolio Deployment with GitHub Pages

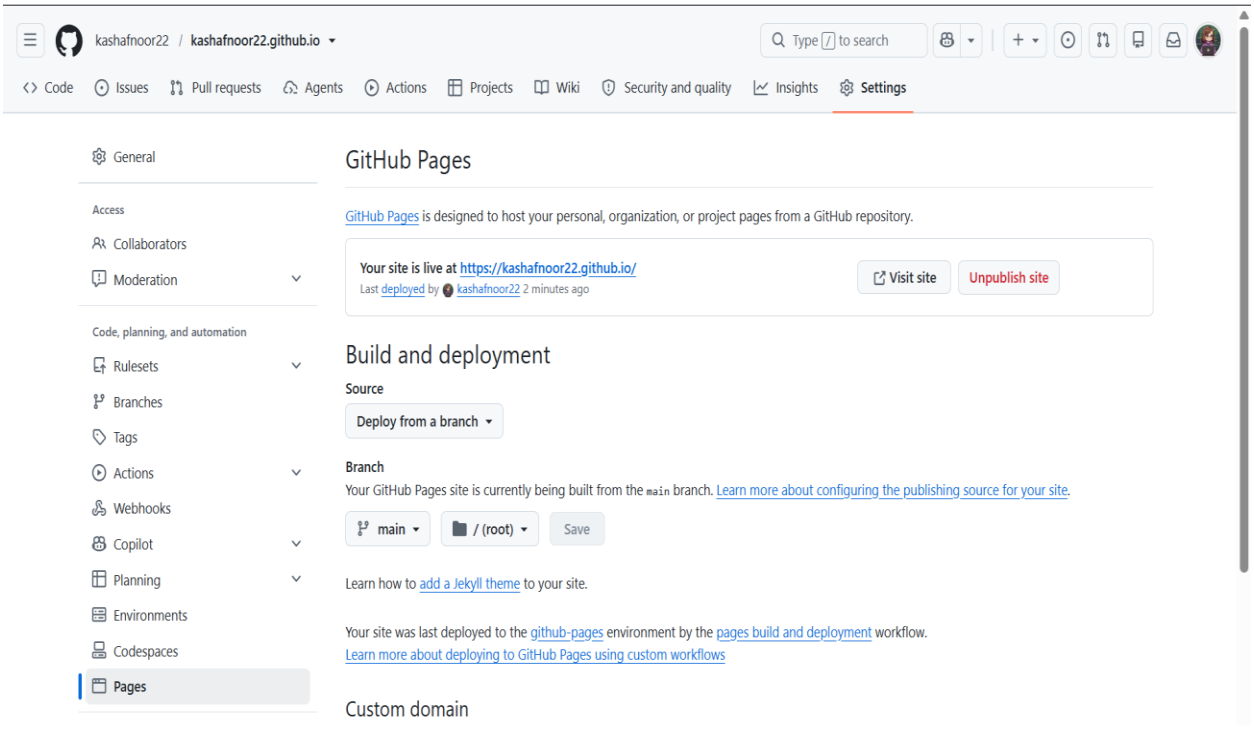
Step 4.1: Portfolio Repository Creation

Description: Created the kashafnoor22.github.io repository on GitHub containing index.html and styles.css.*(Insert Screenshot: task4_pages_repository.png)*



Step 4.2: GitHub Pages Activation

Description: Configured GitHub Pages settings from the main branch root folder and generated the live published site.*(Insert Screenshot: task4_pages_deployment.png)*



Step 4.3: Live Portfolio Verification

Description: Verified the live portfolio website running at https://kashafnoor22.github.io/ displaying all required CV sections.*(Insert Screenshot: task4_portfolio_live.png)*

